

See It Fast! Quantities to Five

Answer Key

Kindergarten–Grade 1

Quick Answers

- | | |
|--|--|
| 1. 3 | = 0 |
| 2. 5 | 6. first part = 3, second part = 2, total = 5 |
| 3. how many dots = 4, how many empty boxes = 1 | 7. card A = 4, card B = 5, comparison symbol = < |
| 4. first part = 2, second part = 2, total = 4 | 8. first part = 4, second part = 1, total = 5, — |
| 5. how many dots = 5, how many empty boxes | |
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What is verified

7 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 8. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1

K.CC.B.4 – *problems 1, 2, 4, 6, 7, 8*

K.CC.B.5 – *problems 3, 5*

K.CC.C.6 – *problem 7*

Difficulty 1–4 of 5 across 19 tagged checks.

- 1.** Diagonal pattern of dots — how many?

Solution: The diagonal line of dots is a pattern to know on sight: one dot in each corner-to-corner step. It shows 3

- 2.** Game-die pattern — how many?

Solution: Four corners and one in the middle — the X pattern from a die. Children who know the pattern name it without counting: 5

- 3.** Five-frame with one empty box: how many dots, how many empty boxes?

Solution: The frame holds five boxes. It is almost full — just one box has no dot, so there is 1

empty box.

Almost-full means one less than five: $5 - 1 = 4$, so the dots are 4

- 4.** Square pattern: write the parts and the total.

Solution: The square splits into a top pair and a bottom pair. The first part is
 The second part is
 Two and two make $2 + 2 = 4$, so the total is

2
2
4

5. Full five-frame: how many dots, how many empty boxes?

Solution: A full frame is the picture of five — no counting needed:
 Every box has a dot, so the empty boxes number

5
0

6. Domino with two halves: write the parts and the total.

Solution: Left half shows the triangle pattern:
 Right half shows the pair:
 Three and two make $3 + 2 = 5$, so the total is

3
2
5

7. Card A (square pattern) and card B (die pattern): numbers, then compare.

Solution: Card A is the square of four:
 Card B is the X of five:
 Four comes before five when counting, so

4
5
$4 < 5$

8. Challenge: write the two parts you see and the total, then tell how you saw it.

Solution: The square of four sits on the left:
 One extra dot sits beside it:
 Four and one more make $4 + 1 = 5$, so the total is

4
1
5

Telling how (instructor-judged): accept any parts-based seeing — “I saw the square of 4 and 1 more,” or “4 and 1 make 5.” Counting all five dots one by one gets the number but does not show the see-it-fast skill; invite the child to look for the square inside the pattern.